Fe-C Problem 9 | Problems T4_PhaseDiagrams II.pdf #9 | Fe-C · secondary cementite

A 3 % C Cast Iron from 1146 to 800 °C

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 3

9. Examine closely the phase diagram Fe-C answer the following questions. a) Calculate the percentage of C in an alloy that has 64.7 % of austenite and 35.5 % of cementite at 800 ºC. What kind of alloy is it? b) What microconstituents does it have at 1146ºC? Calculate the quantity of each one. c) How do microconstituents change when cooling down from 1146 ºC to 800 ºC?

Official Solution: a) C_0 **= 3%; hypoeutectic cast iron; b)** $w_{\gamma primary}$ **= 60.2%;** $w_{ledeburite}$ **= 39.8%;** $w_{Fe_3C\ ledeburite}$ **= 18.8%;** $w_{\gamma ledeburite}$ **= 21%;** $w_{\gamma total}$ **= 81.2%; c) Fe₃C precipitates (secondary Fe₃C);** $w_{Fe_3C\ secondary}$ **= 16.3%**

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium, Fe-Fe3C diagram [Session 9 Slide 4].
2. **Reading of the figure:** at 800 °C the A_{cm} line is at about 1.0 % C (a straight line from (0.76, 727) to (2.14, 1147) gives $0.76 + 73 \times 1.38/420 = 1.00$).
3. The data of the statement add up to 100.2 %; each fraction is used on its own to get C_0, and the two results are compared.

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C). Tie lines: 800 °C (γ 1.0 / Fe3C 6.67), 1147 + ΔT (γ 2.14 / L 4.30), 1146 °C (γ 2.14 / Fe3C 6.67). The lever rule is solved for C_0 in (a) and applied in (b) and (c); microconstituents as in Session 7 Slide 32.

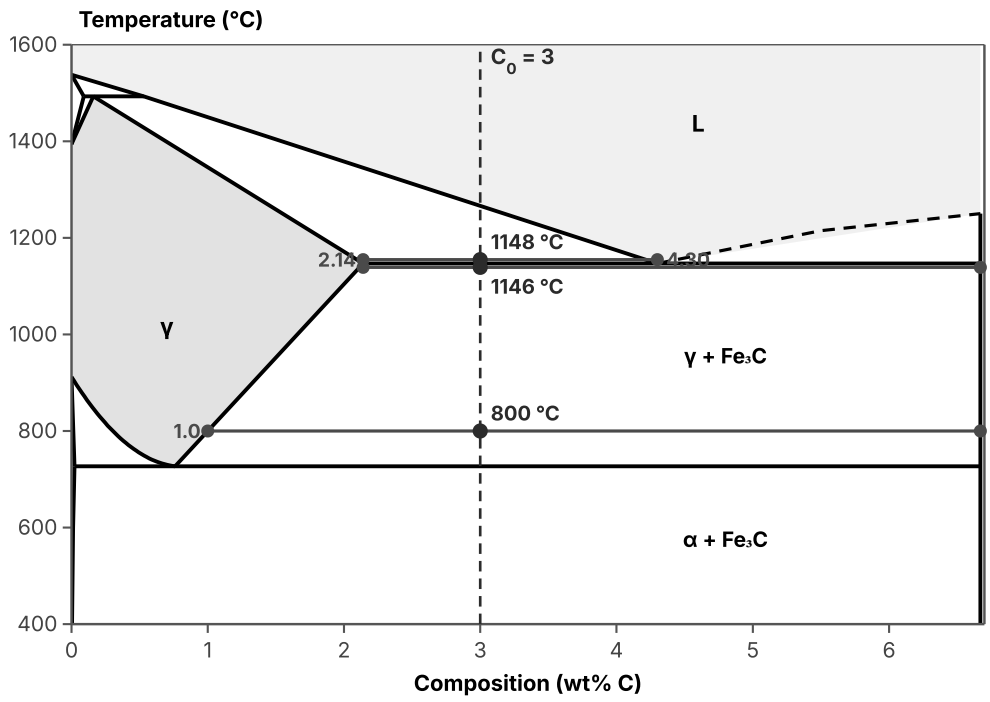


Figure 37. Fe-3 % C: γ + L just above 1147 °C, primary γ + ledeburite just below it, γ (about 1.0 % C, a reading) + Fe_3C at 800 °C.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Composition From the austenite: $W_\gamma = \dfrac{6.67 - C_0}{6.67 - 1.0} = 0.647 \Rightarrow C_0 = 6.67 - W_\gamma\,(6.67 - 1.0) = 6.67 - 0.647 \times 5.67 = 6.67 - 3.668 = 3.00$ % C.

From the cementite: $W_{\mathrm{Fe_3C}} = \dfrac{C_0 - 1.0}{5.67} = 0.355 \Rightarrow C_0 = 1.0 + W_{\mathrm{Fe_3C}} \times 5.67 = 1.0 + 0.355 \times 5.67 = 1.0 + 2.013 = 3.01$ % C.

Both give C_0 **≈ 3.0 % C**: $2.14 < 3.0 < 4.30$, a **hypoeutectic cast iron**.

b) Microconstituents at 1146 °C Just above 1147 °C (γ 2.14 / L 4.30; length 2.16):

$$W_{\gamma,p} = \frac{4.30 - 3.0}{2.16} = \frac{1.30}{2.16} = 0.602 \qquad W_L = \frac{3.0 - 2.14}{2.16} = \frac{0.86}{2.16} = 0.398$$

At 1146 °C the liquid has become **ledeburite** (0.398) and the **primary austenite** remains (0.602). Phases at 1146 °C (γ 2.14 / Fe3C 6.67; length 4.53):

$$W_{\mathrm{Fe_3C}} = \frac{3.0 - 2.14}{4.53} = \frac{0.86}{4.53} = 0.190 \qquad W_{\gamma,\mathrm{total}} = 1 - 0.190 = 0.810$$

All the cementite is in the ledeburite (0.190), so the austenite of the ledeburite is $0.398 - 0.190 = 0.208$.

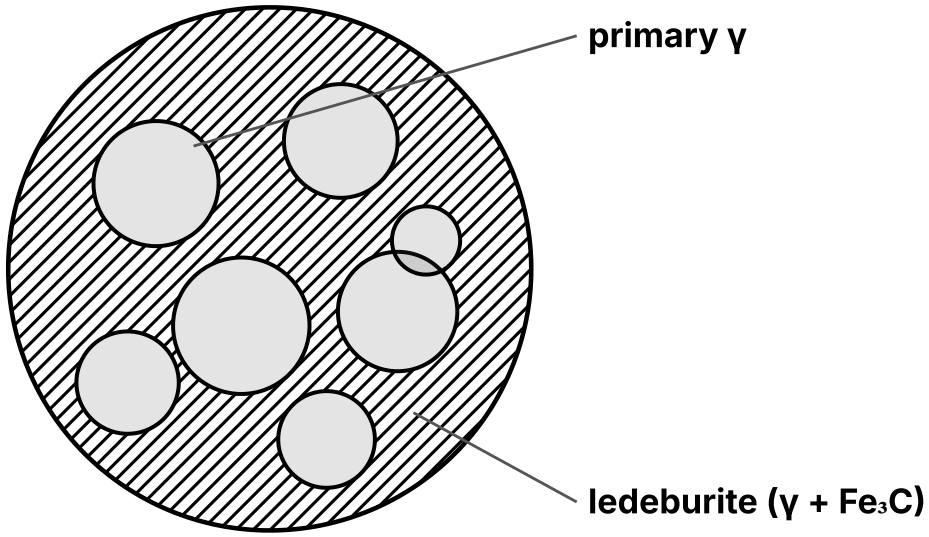


Figure 38. Fe-3 % C just below 1147 °C: primary austenite (60 %) and ledeburite (40 %).

c) From 1146 to 800 °C The austenite (primary and eutectic) loses carbon along A_{cm}, from 2.14 to 1.0 % C, and the excess precipitates as **secondary cementite** [Session 9 Slide 11]. At 800 °C:

$$W_{\mathrm{Fe_3C}}(800) = \frac{3.0 - 1.0}{5.67} = \frac{2.0}{5.67} = 0.353 \qquad W_{\mathrm{Fe_3C,sec}} = 0.353 - 0.190 = 0.163$$

PHASE 4 · CHECKS & INTERPRETATION

- Ledeburite make-up: $0.190/0.398 = 0.477$ Fe3C, as in Worked example 14 .
- At 800 °C: γ $1 - 0.353 = 0.647$ – exactly the 64.7 % of the statement .
- **Limiting cases** (fractions dimensionless, compositions in wt% C): at 800 °C the austenite fraction is 1 at $C_0 = 1.0$ and 0 at $C_0 = 6.67$; 3.0 % C lies about a third of the way along, hence 0.647 of austenite.

Check against the official solution

Agrees with the key: 3 %, hypoeutectic cast iron, 60.2 %, 39.8 % and 16.3 %.

DISCREPANCY WITH THE OFFICIAL SOLUTION

With 6.67 % C the cementite of the ledeburite is 19.0 % (key 18.8 %), its austenite 20.8 % (key 21 %) and the total austenite 81.0 % (key 81.2 %). The key uses 6.7 % C for cementite, which gives 18.86 %, so its 18.8 is also truncated. Differences of 0.2 percentage points at most; the computed values are kept.